

MAAT v1.6-T: The Triage Instantiation

Structural Search Geometry as a Pre-Inference Layer

Christof Krieg

MAAT Project – Independent Research

<https://maat-research.com>

2026

Abstract

MAAT v1.6 defines structural search geometry: a framework in which candidate states, paths, and transitions are ranked by bounded structural support coordinates before full inference, proof, simulation, or solving is committed. This document specifies the *triage* instantiation, denoted v1.6-T.

The v1.6 layer has two faces. The *measure* face treats paths as objects of a structural path measure, with quasipotentials and transition rates. The *policy* face treats the same geometry as a compute-allocation rule: given a search state, which candidate deserves the next unit of compute? The two are connected by a finite-horizon marginal policy

$$\pi(c \mid n) \propto \exp[-\beta h_{\text{MAAT}}^{(h)}(c)].$$

At $h \rightarrow \infty$, the heuristic approaches a quasipotential-like cost-to-go. At $h = 0$, it becomes greedy structural triage.

The central guardrail is goal-blindness: F_{MAAT} measures structural coherence, not progress toward a target. Therefore pure structural triage is not a complete search policy. The specification defines two sanctioned usage modes: Mode F, a feasibility prior, and Mode A, a combined acquisition rule

$$A(c) = \text{prog}(c) - \tau h_{\text{MAAT}}^{(h)}(c).$$

This document also states the honest baseline bar and predeclares the first four obligations T1–T4. It is a specification, not evidence. Paper 62 implements the first T1 benchmark.

1 Purpose

MAAT v1.6 reframes structural selection as search geometry. Earlier MAAT layers ranked states or trajectories by structural support. The triage instantiation asks a more operational question:

Which candidates are structurally coherent enough to deserve compute before full inference, proof, or solving is run?

This matters because expensive search is often bottlenecked not by the final validation criterion but by allocation. A SAT solver chooses a branch before it knows whether the branch is fruitful. A proof assistant chooses a lemma path before it knows whether the proof closes. A model search chooses a region before evaluating a full likelihood. A scientific programme chooses which hypothesis to test before the complete experiment exists.

MAAT v1.6-T is therefore not a truth criterion. It is a bounded, falsifiable pre-inference ranking layer.

2 Two Faces of v1.6

The v1.6 path-selection layer admits two complementary instantiations.

Measure instantiation. The measure face defines path probabilities, structural path actions, and quasipotential-like quantities. In schematic form,

$$P[\gamma] \propto \exp[-F_{\text{path}}[\gamma]],$$

where γ is a path through a structural configuration space. This face is appropriate when one studies transition rates, basin escape, relaxation paths, or physical/dynamical ensembles.

Policy instantiation. The triage face defines compute-allocation rules over candidate continuations. It is appropriate when one has a search state n and a finite candidate set $\mathcal{C}(n)$, but cannot afford to evaluate every continuation fully. The output is not a probability of physical existence. It is a ranking or sampling policy over the next candidates to try.

The two faces are the same geometry at different horizons. The measure face asks for the full path structure. The triage face asks for a finite-horizon approximation to that structure.

3 Search Arena

A search problem is represented as a tuple

$$(\mathcal{N}, \mathcal{C}, \text{cost}, \text{goal}),$$

where:

- $\mathcal{N}$ is the space of search states,
- $\mathcal{C}(n) \subset \mathcal{N}$ is the candidate-continuation set available at state n ,
- $\text{cost}(n \rightarrow c)$ is the compute cost of expanding candidate c ,
- goal is the domain validation criterion or objective.

The dynamics layer defines what candidates are admissible. In SAT, this is the solver transition rule. In proof search, it is the proof calculus. In model fitting, it is the permitted parameter update or evaluation rule. The MAAT triage layer does not override admissibility. It ranks only candidates already supplied by the domain dynamics.

4 Structural Heuristic

Each candidate c is mapped to structural supports evaluated on its associated local configuration Φ_c :

$$c \mapsto (H, B, S, V)[\Phi_c] \mapsto R_{\text{rob}}[\Phi_c] \mapsto F_{\text{MAAT}}[\Phi_c].$$

The standard v1.2.1 robustness closure is retained:

$$R_{\text{resp}} = (HBV)^{1/3}, \tag{1}$$

$$R_{\text{rob}} = \min\left(R_{\text{resp}}, (HBSV)^{1/4}\right). \tag{2}$$

The bounded multi-support score is

$$F_{\text{MAAT}}[\Phi_c] = - \sum_a \lambda_a \log(\varepsilon + \Gamma_a[\Phi_c]) - \lambda_{\text{cl}} \log(\varepsilon + R_{\text{rob}}[\Phi_c]). \tag{3}$$

For search, this immediate score generalizes to a finite-horizon structural heuristic:

$$h_{\text{MAAT}}^{(h)}(c) = \min_{\gamma: c \rightarrow \cdot, |\gamma| \leq h} F_{\text{path}}[\gamma], \quad (4)$$

estimated greedily, by sampling, or by a domain-local rollout. At $h = 0$, this reduces to the immediate candidate score:

$$h_{\text{MAAT}}^{(0)}(c) = F_{\text{MAAT}}[\Phi_c].$$

5 Unification Equation

The triage policy is the finite-horizon marginal of the v1.6 path measure:

$$\pi(c \mid n) \propto \exp[-\beta h_{\text{MAAT}}^{(h)}(c)] \quad (5)$$

where β controls exploration versus exploitation.

Two limits anchor the construction:

$$h \rightarrow \infty : \quad h_{\text{MAAT}}^{(h)}(c) \rightarrow V_q(c \rightarrow \text{goal basin}), \quad (6)$$

$$h = 0, \beta \rightarrow \infty : \quad \pi(c \mid n) \rightarrow \text{greedy ranking by } F_{\text{MAAT}}[\Phi_c]. \quad (7)$$

Thus h_{MAAT} plays the same structural role as a heuristic cost-to-go in A* search, while V_q is the full cost-to-go it approximates.

6 Goal-Blindness

The critical limitation is fundamental:

F_{MAAT} measures coherence, not progress.

Structural coherence is not the same as solving the problem. A coherent branch can be a dead end. An incoherent branch can be one step away from a proof, satisfying assignment, or good fit. Therefore pure structural triage must fail in domains where coherence and promise decouple.

This is not a technical detail. It is the main guardrail of v1.6-T. The structural layer must either be used as a feasibility prior or combined with a domain progress term.

7 Two Sanctioned Usage Modes

7.1 Mode F: Feasibility Prior

Mode F uses structural selection only to filter or down-weight broken candidates:

$$c \in \mathcal{C}(n) \quad \mapsto \quad w_F(c) = \exp[-\beta h_{\text{MAAT}}(c)].$$

A domain heuristic then ranks the surviving candidates. Mode F is cheap, safe, and limited in upside. It should be preferred when structural support is known to be weakly coupled to target progress.

7.2 Mode A: Combined Acquisition

Mode A is the stronger policy. It combines domain progress with structural risk:

$$A(c) = \text{prog}(c) - \tau h_{\text{MAAT}}^{(h)}(c) \quad (8)$$

where $\text{prog}(c)$ is a domain progress estimate and $\tau \geq 0$ trades progress against structural risk.

Examples of $\text{prog}(c)$ include:

- unit-propagation gain or VSIDS activity in CDCL,
- expected likelihood improvement in model fitting,
- lemma-depth or proof-state reduction in proof search,
- validation gain in machine-learning configuration search.

Mode A is structurally analogous to acquisition functions in Bayesian optimization, but h_{MAAT} acts as a structural risk penalty rather than as a posterior uncertainty term.

8 Budget Allocation

Given a finite budget K , candidates are sampled or ranked by

$$\pi_A(c \mid n) \propto \exp[\beta A(c)].$$

The primary evaluation metric is not rank correlation. It is compute-to-solution regret:

$$\Delta C_\pi = C_\pi - C_{\text{baseline}},$$

where C_π is the compute used by policy π . Negative regret means that the triage policy used less compute than the baseline on the same task.

9 Honest Baseline Bar

In deployment domains, shuffled nulls are not enough. The triage layer must compete with strong domain heuristics.

Domain	Mandatory baselines
SAT/CDCL	VSIDS, LRB/CHB, random branching, restart portfolios
Model fitting	Expected Improvement, UCB, uniform scans
Proof search	depth/size heuristics, premise-selection learners
Any domain	shuffled- λ null, F_{MAAT} -only, progress-only

A triage layer that beats shuffled nulls but loses to VSIDS is a negative result for SAT branching and must be reported as such.

10 What Is New

The acquisition pattern is not new. Softmax exploration is not new. A* heuristics are not new. The candidate novelty of v1.6-T consists of two claims.

N1: Specific structural risk form. The bounded multi-support risk score

$$(H, B, S, V, R_{\text{rob}}) \mapsto h_{\text{MAAT}}$$

is claimed to capture structural risk that scalar alternatives miss. This claim is not established by definition. It must be tested against domain baselines.

N2: Transferability of structural modes. Some structural support modes, especially low-frequency connectivity modes, may transfer across domains better than full scalar scores. This claim is suggested by Papers 52–53 but remains a policy-level hypothesis.

If neither N1 nor N2 survives benchmarking, v1.6-T reduces to known acquisition-function practice with new notation and should be revised or retired.

11 Predeclared Obligations

The triage evidence registry begins empty. The following obligations are predeclared:

T1: SAT branching pilot. Implement Mode A inside a CDCL solver on random and crafted instances. Report compute-to-solution regret against VSIDS and, in later versions, LRB/CHB. Predeclare β, τ, h , seeds, and instance families. Paper 62 implements the first T1 pilot.

T2: Mode ablation. On the same instances, compare Mode F, Mode A, progress-only, and F_{MAAT} -only. If progress-only matches Mode A, the structural term adds no compute value at that policy point.

T3: Frozen-weight triage transfer. Freeze a calibrated structural policy in one domain and apply it to another candidate-ranking task without retuning. This extends Paper 52 from score transfer to policy transfer.

T4: Consistency with the measure face. On a small enough arena to compute the quasipotential V_q exactly, verify whether $h_{\text{MAAT}}^{(h)}$ orderings approach V_q orderings as h grows. This tests the unification equation empirically.

Falsification rule. v1.6-T is weakened or abandoned if T2 shows the structural term redundant across domains, or if T1 regret is consistently worse than the relevant domain baseline across tested (β, τ, h) .

12 Two Closure Regimes

The distinction between physical/algorithmic closure and normative closure remains important.

Physics and algorithms. High- F_{MAAT} candidates are deprioritized, not forbidden. The solver may still explore them under stochasticity, backtracking, or explicit exploration.

Normative AI systems. Candidates violating hard R -constraints are removed from $\mathcal{C}(n)$ before ranking. An aligned agent does not merely assign low priority to a truthfulness- or safety-violating action. It does not include it as an admissible continuation.

In search, incoherence is deprioritized. In alignment, hard R -violations are not in the candidate set.

13 Relation to Paper 62

Paper 62 is not the full v1.6-T theory. It is the first T1 benchmark. It instantiates Mode A at:

$$(\beta, \tau, h) = (8.0, 0.38, 0)$$

inside a transparent CDCL solver and compares compute-to-solution regret against VSIDS.

The result is mixed: the global $h = 0$ structural penalty does not beat the progress-only ablation, but family-level structure appears. This motivates the next v1.6-T variant:

$$A_{\text{gated}}(c) = \text{prog}(c) - g(\mathcal{F})\tau h_{\text{MAAT}}(c),$$

where $g(\mathcal{F})$ activates structural steering only when the formula contains exploitable low-frequency structural modes.

14 Conclusion

MAAT v1.6-T is the policy face of structural search geometry. It defines a pre-inference triage layer, not a truth criterion. Its central equation is

$$\pi(c \mid n) \propto \exp[-\beta h_{\text{MAAT}}^{(h)}(c)],$$

and its deployable form is either Mode F, a feasibility prior, or Mode A, a combined acquisition rule.

The decisive scientific requirement is compute accountability:

the structural ranking must earn its compute.

Paper 62 begins that test. The full programme remains open and falsifiable.

References

- [1] P. E. Hart, N. J. Nilsson, and B. Raphael, “A formal basis for the heuristic determination of minimum cost paths,” *IEEE Transactions on Systems Science and Cybernetics* **4**, 100–107 (1968).
- [2] M. W. Moskewicz, C. F. Madigan, Y. Zhao, L. Zhang, and S. Malik, “Chaff: Engineering an efficient SAT solver,” in *Proceedings of the 38th Design Automation Conference*, 530–535 (2001).
- [3] A. Biere, M. Heule, H. van Maaren, and T. Walsh (eds.), *Handbook of Satisfiability*, IOS Press, Amsterdam (2009).
- [4] S. Pressé, K. Ghosh, J. Lee, and K. A. Dill, “Principles of maximum entropy and maximum caliber in statistical physics,” *Reviews of Modern Physics* **85**, 1115–1141 (2013).
- [5] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. de Freitas, “Taking the human out of the loop: A review of Bayesian optimization,” *Proceedings of the IEEE* **104**, 148–175 (2016).
- [6] C. Krieg, “Structural Search Geometry: A MAAT v1.6 Framework Note with a Toy Search Benchmark,” preprint (2026).
- [7] C. Krieg, “Mode-A CDCL Branching: First Policy-Level Test of MAAT v1.6-T Structural Search,” preprint (2026).